Date: 25/03/2026

Ex-10
Explore Antivirus Detection Techniques

Aim:

To study and understand the different techniques used by antivirus software to detect malicious programs and protect systems from threats, including static, dynamic, and heuristic detection methods.

Procedure:

- Install antivirus software (e.g., Windows Defender, ClamAV) on the target system.
- Update the antivirus database to the latest version.
- Scan a sample file or folder using the antivirus tool.
- Observe how the antivirus detects threats using signature-based, heuristic, and behavior-based methods.
- Analyze the scan report, identified threats, and AV fingerprinting techniques.

Task-1:

Task 1 Introduction

Welcome to Intro to AV

Antivirus (AV) software is one of the essential host-based security solutions available to detect and prevent malware attacks within the end-user's machine. AV software consists of different modules, features, and detection techniques, which are discussed in this room.

As a red teamer or pentester, it is essential to be familiar with and understand how AV software and its detection techniques work. Once this knowledge is acquired, it will be easier to work toward AV evasion techniques.

Learning Objectives

- What is Antivirus software?
- Antivirus detection approaches
- Enumerate installed AV software in the target machine
- Test in a simulated environment

Room prerequisites

- General knowledge of host-based detection solutions; check The Lay of the Land room for more information.
- General experience with Hashing crypto; check the Hashing - Crypto 101 room for more information.
- Basic knowledge of Yara Rules; check the THM Yara room for more information.

Answer the questions below

Let's get started!

No answer needed

✓ Correct Answer

Task-2:

McAfee Associates, Inc. started the first AV software implementation in 1987. It was called "VirusScan," and its main goal at that time was to remove a virus named "Brain" that infected John McAfee's computer. Later, other companies joined in the battle against viruses. AV software was called scanners, and they were command-line software that searched for malicious patterns in files.

Since then, things have changed. AV software nowadays uses a Graphical User Interface (GUI) to perform scans for malicious files and other tasks. Malware programs have also expanded in scope and now target victims on Windows and other operating systems. Modern AV software supports most devices and platforms, including Windows, Linux, macOS, Android, and iOS. Modern AV software has improved and become more intelligent and sophisticated, as they pack a bundle of versatile features, including Antivirus, Anti-Exploit, Firewall, Encryption tool, etc.

We will be discussing some AV features in the next task.

Answer the questions below

What does AV mean?

Antivirus — ✓ Correct Answer

Which PC Antivirus vendor implemented the first AV software on the market?

McAfee — ✓ Correct Answer

Antivirus software is a _____-based security solution.

Host — ✓ Correct Answer

Task-3:

Other common features

The following are some common features found in AV products:

- A self-protection driver to guard against malware attacking the actual AV.
- Firewall and network inspection functionality.
- Command-line and graphical interface tools.
- A daemon or service.
- A management console.

Answer the questions below

Which AV feature analyzes malware in a safe and isolated environment?

Emulator — ✓ Correct Answer

An _____ feature is a process of restoring or decrypting the compressed executable files to the original.

unpacker — ✓ Correct Answer

Read the above to proceed to the next task, where we discuss the AV detection techniques.

No answer needed — ✓ Correct Answer

Task-4:

100%

Task 1 Introduction

Welcome to Intro to AV

You can also connect via RDP, make sure you deploy the AttackBox or connect to the VPN.

Use the following credentials below.

Machine IP: `MACHINE_IP` Username: `thm` Password: `TryHackM3`

Connect to the VM via the RDP client

```
user@machine$ xfreerdp /v:MACHINE_IP /u:thm /p:TryHackM3
```

Answer the questions below

Once you've deployed the VM, it will take a few minutes to boot up. Then, progress to the next task!

No answer needed ✓ Correct Answer

Task-5:

The output shows we improved our Yara rule to reduce the false-positive results. That was a simple example of how AV software works. Thus, AV software vendors work hard to fight against malware and improve their products and database to enhance the performance and accuracy of results.

The drawback of the signature-based detection is that files will have a different hash value if the binary is modified. Therefore, it is easy for someone to bypass signature-based detection techniques if they know what AV software looks for and how to analyze binaries, as shown in later rooms.

Answer the questions below

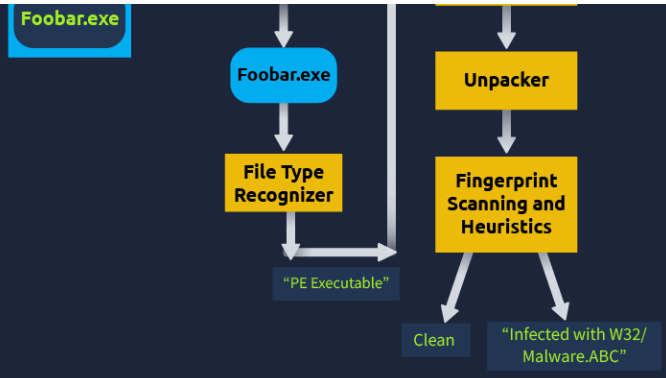
What is the `sigtool` tool output to generate an MD5 of the `AV-Check.exe` binary?

f4a974b0cf25dca7fbce8701b7ab3a88:6144:AV-Check.exe ✓ Correct Answer

Use the strings tool to list all human-readable strings of the AV-Check binary. What is the flag?

THM{Y0uC4nC-5tr16s} ✓ Correct Answer

Task-6:



In the diagram, you can see a suspicious `Foobar.zip` file is passed to AV software to scan. AV software recognizes that it is a compressed file (.zip). Since the software supports .zip files, it applies an un-archiver feature to extract the files (`Foobar.exe`). Next, it identifies the file type to know which module to work with and then performs a PE parsing operation to pull the binary's information and other characteristic features. Next, it checks whether the file is packed; if it is, it unpacks the code. Finally, it passes the collected information and the binary to the AV engine, where it tries to detect if it is malicious and gives us the result.

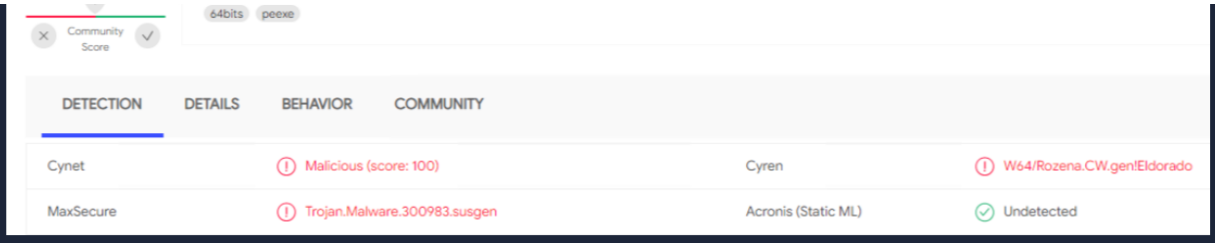
Answer the questions below

Which detection method is used to analyze malicious software inside virtual environments?

Dynamic Detection

✓ Correct Answer

Task-7:



Note: if you try to submit a file to the VirusTotal website, it may give you a different result. Keep in mind that VirusTotal shares submission reports with the Antivirus vendors to improve their AV detection engines, including false-positive results.

Answer the questions below

For the C# AV fingerprint, try to rewrite the code in a different language, such as Python, and check whether VirusTotal flag it as malicious.

No answer needed

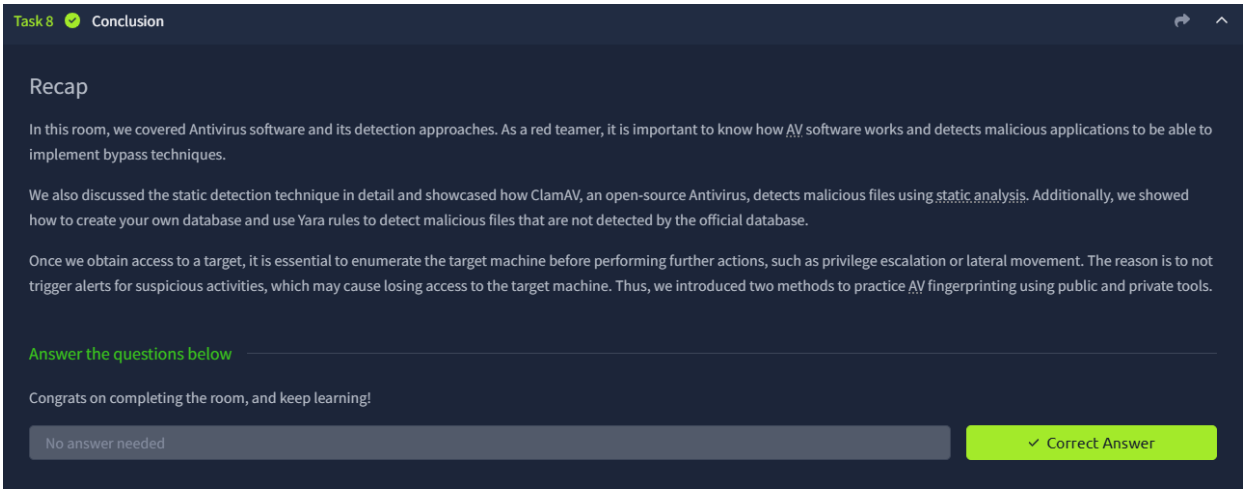
✓ Correct Answer

Read the Above!

No answer needed

✓ Correct Answer

Task-8:



Result:

Antivirus detection techniques were successfully explored through static analysis (MD5 hashing, YARA rules, ClamAV), dynamic detection (emulation, sandboxing), and heuristic methods. The experiment demonstrated how antivirus software effectively identifies both known and unknown threats using multiple detection approaches, ensuring better system security and protection from malware.